MECHANICAL ENGINEERING & PHYSICS

Muhammad Maalik

Engineering portfolio

From an idea to something built.

Robotics, flight, space systems, and engineering analysis. A selection of hardware, simulations, and the decisions connecting them.

muhammadmaalik.github.io | umermaalik24@gmail.com

Stony Brook University

B.S. Mechanical Engineering & Physics / August 2025 - May 2030

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PROJECT 01 / JULY 2026 - PRESENT

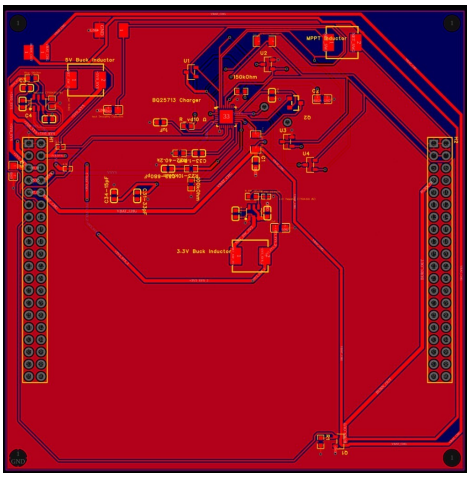
PsiSat

A 1U gamma-ray burst nanosatellite

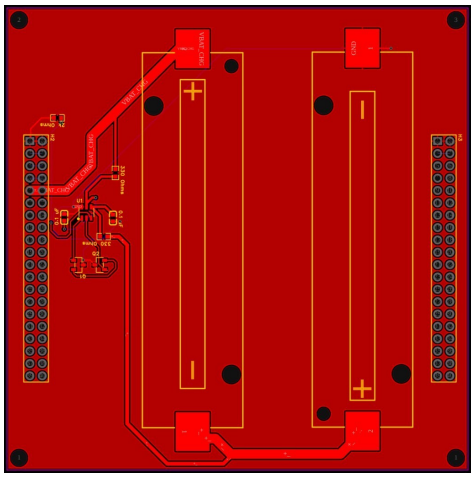
Design study / proposal

A student CubeSat proposal to detect short flashes of high-energy light from orbit, with a compact aluminum shell and six planned electronics layers.

TOOLS / Fusion 360, PCB design, detector integration



Stage 4: power board layout.



Stage 5: battery board layout.

Purpose

A scintillator converts incoming gamma rays into light near 550 nm. A sensor reads each flash and a small computer records its timing.

My contribution

I started the project, recruited the team, and wrote the NASA CubeSat Launch Initiative proposal. The shell is modeled in Fusion 360. Jackson Boyle and Christian Tumanda are also on the proposal.

Packaging & status

The target envelope is about 10 x 10 x 11 cm, with mass below 1.33 kg. Six electronics layers are planned. Stage 4 is the power board and Stage 5 holds the batteries; four other boards remain in design. PsiSat has not launched.

View project repository

PROJECT 02 / DECEMBER 2025 - PRESENT

3D-printed jet

Airframe, propulsion & flight-control development

In development

A custom printed airframe with twin 80 mm electric ducted fans, developed around structural packaging, aerodynamics, and a target speed above 70 km/h.

TOOLS / Fusion 360, Inventor, Ansys Mechanical, Ansys Fluent, PyTorch



CAD model of the airframe, fans, and tail.



Printed wing panels and battery bay.

Airframe & integration

The airplane is drawn in Fusion 360 and printed in separate parts. The wing halves close over a battery bay, while two 80 mm electric ducted fans sit side by side at the rear.

Analysis & control

Structural and flow analysis inform the design. Thrust vectoring and onboard neural-network flight and landing control remain development goals. The 70 km/h figure is a design target, rather than a reported flight measurement.

Explore the model

The original interactive aircraft model and flight demonstration remain available on the main website.

PROJECT 03 / SUMMER INTERNSHIP, 2026

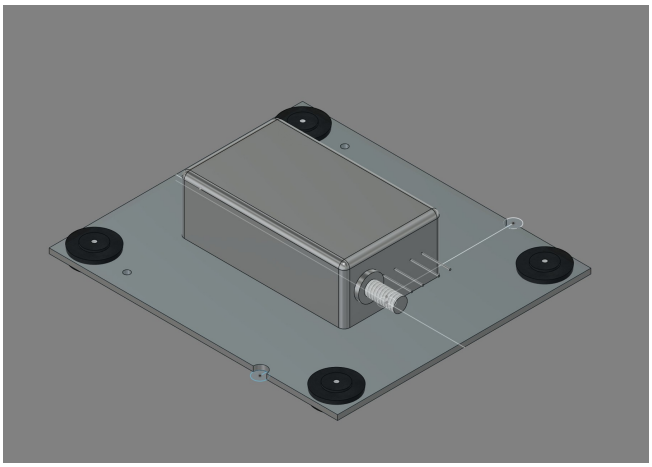
Vibration isolation

Crystal oscillator mounting study

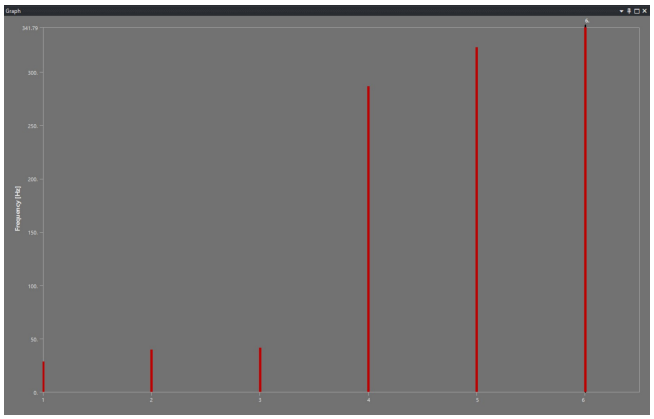
Engineering study

A modal study of a 10 MHz crystal oscillator on four isolators, identifying the frequency ranges where its mounts amplify or reduce vibration.

TOOLS / Ansys Mechanical, modal analysis



Oscillator plate and four isolators.



Ansys results for modes 1-6.

The assembly

At East End Assemblies, I studied a Wenzel 501-27521-11 oscillator mounted on a plate with four J-3112-2 isolators.

The analysis

I ran modal analysis in Ansys Mechanical and examined the first six natural frequencies. Low modes were near 30-50 Hz, where bench motion can be amplified. Isolation began to help above roughly 70 Hz; the mounts stiffened at higher frequencies.

Engineering purpose

The study described where the mounting system helps and where it does not before installation. These findings apply to the studied assembly and model.

PROJECT 04 / SEPTEMBER 2025 - PRESENT

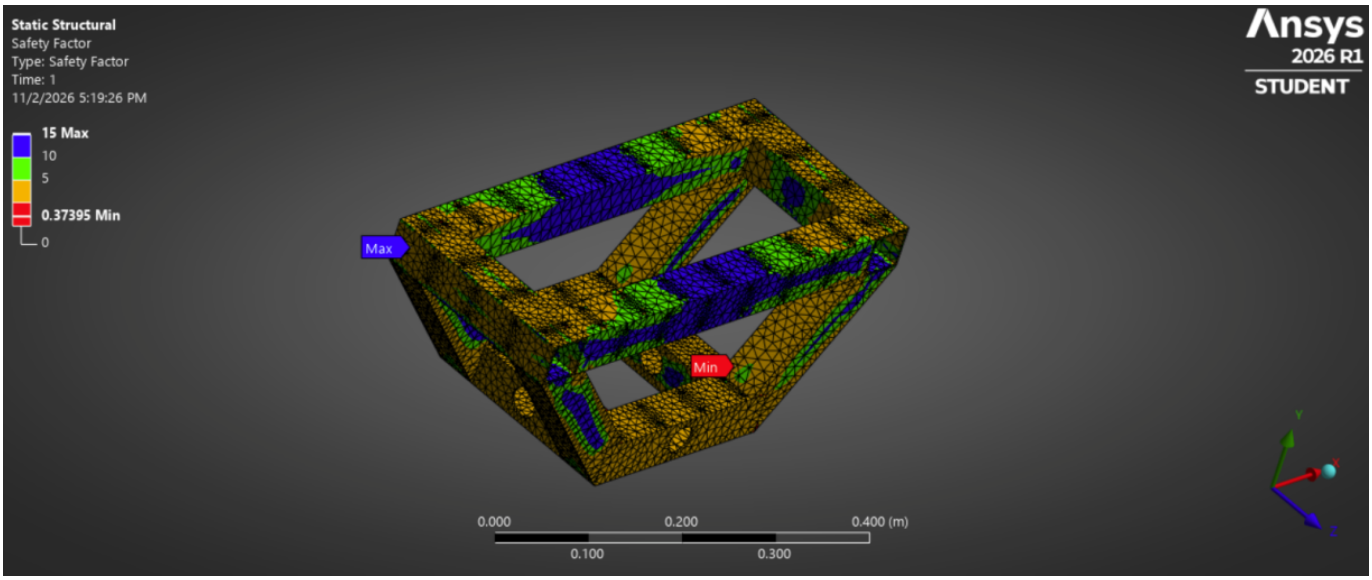
Rover structural analysis

Frame design, load cases & weight reduction

FEA lead / Stony Brook Robotics

Design and analysis of the URC rover frame, using static and impact cases to guide structural changes before fabrication.

TOOLS / SolidWorks, Ansys Mechanical, LS-DYNA



Static structural safety-factor plot (2 Nov 2026); displayed minimum 0.37395.

My contribution

I design the URC rover frame in SolidWorks and evaluate it in Ansys Mechanical. LS-DYNA supports short impact cases. Successive load-case iterations reduced the rover mass by 6 kg.

Top-frame study

The September 2026 pass modeled aluminum 6061-T6 members with 5 mm joint connections. Loads of 100 N, 150 N, and 150 N were applied with selected faces fixed. This initial pass used the default mesh.

Design response

Peak deformation on the middle member pointed to the layout as the area to improve. Proposed changes included additional supports, filleted trusses or L-bars, and evaluation of 2 x 2 inch tubing.

PROJECT 05 / SBU HACKS 2026

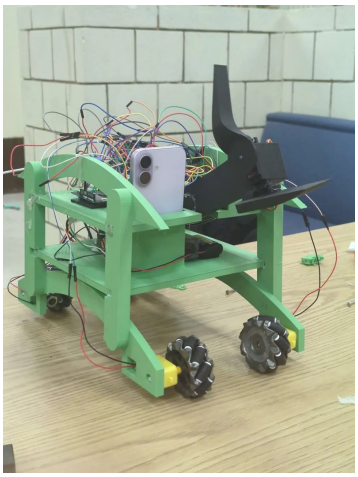
Batman

Autonomous trash-pickup robot

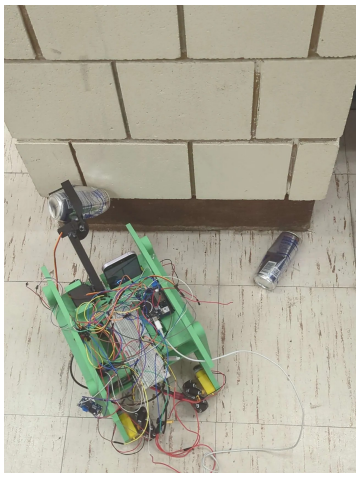
Team prototype

A low-cost robot designed to find and collect campus litter, bringing computer vision, omnidirectional motion, and a two-servo claw into one hardware prototype.

TOOLS / Arduino UNO Q, Ultralytics YOLO11, Python, mecanum drive, TCP / Wi-Fi



The assembled prototype: mecanum base, iPhone camera, and two-servo claw.



Claw and chassis during a can-pickup test.

The idea

Litter collects under benches, beside bins, and along campus walkways. Our team built Batman to explore a small, affordable robot that can reach those spots and pick it up.

See, decide, move, grab

An iPhone streams video to a Windows laptop running a trash-trained YOLO11 model on an RTX 5060. Drive and arm commands travel over TCP on a shared hotspot to the UNO Q. Four motors and two HW-166 (TB6612) drivers power the mecanum base; lift and gripper servos operate the claw.

Integration & lessons

We tested the motors and servos separately before integrating vision. Stable logic power and a common ground resolved driver issues. The stock Zephyr core lacked Servo.h, so the team used soft 50 Hz servo pulses. A shared hotspot replaced campus Wi-Fi for the robot link.

Current state & next steps

The physical base, arm controller, and GPU detector are built. Reliable pickup remains a tuning task: center on detections, improve the grab sequence, tidy the harness, and extend battery runs. Obstacle sensing and onboard compute are future options.

View project repository

PROJECT 06 / HACKATHON PROJECT

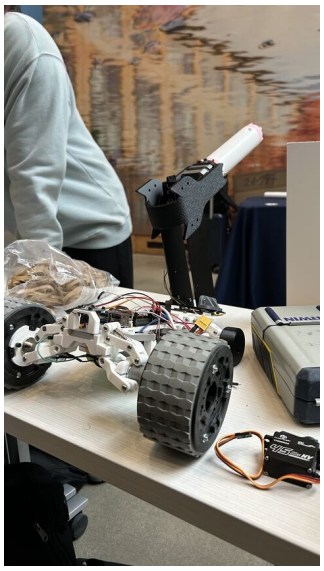
MIT Make - Harvard

Mobile rover & turret prototype

Team prototype

A wheeled rover carrying a small aiming turret, combining a camera-based recognition system with a compact mechanical assembly.

TOOLS / Mechanical integration, camera vision, 3D printing



Rover and turret on the bench.

The prototype

The robot combines a wheeled base, a small turret, and a launcher for short 3D-printed cylinders. Camera-based face recognition triggers the prototype behavior.

Mechanical integration

The mechanical task was to package the launcher, aiming hardware, and wiring on a rover that could still move. The photograph documents the bench prototype.

PROJECT 07 / COMPLETED

Thrust vector control

Compact rocket steering mechanism

Design & simulation

A compact mechanism that steers a rocket by tilting its engine, designed around a roughly 20 x 4 inch, 200 gram vehicle.

TOOLS / Fusion 360, OpenRocket

Packaging & control

Modeled the system in Fusion 360 to fit the actuators, controller, and boards within the rocket. Tilting the engine changes thrust direction to counter disturbances.

Simulation

OpenRocket studies checked system compatibility, center of gravity, and center of pressure to assess the steering design before flight.

PROJECT 08 / JUNE 2024

Hydrogen engine study

Comparative performance & operating constraints

Comparative study

An investigation of electrolysis-derived hydrogen in an internal combustion engine, comparing performance and operating demands with gasoline.

TOOLS / Research, Python, data analysis



Hydrogen engine experiment: gas-generation apparatus, tubing, and engine.



Gas-generation apparatus connected to the battery supply.

Research question

Compared engine behavior and operating overhead for hydrogen and gasoline, focusing on efficiency, speed, and thermal demands.

Reported observations

The study reported 40% higher RPM and 60% higher operating temperature for the hydrogen-fueled engine relative to the gasoline baseline. The thermal result motivated closer attention to material constraints.

Design implication

The temperature result highlights thermal management and material constraints as priorities for subsequent investigation.

EXPERIENCE & TOOLS

Engineering with teams.

Mechanical engineering intern / East End Assemblies

June 2026 - present. CAD and engineering analysis for parts and assemblies, including the oscillator vibration study documented in this portfolio.

FEA lead / Stony Brook Robotics

September 2025 - present. Structural analysis for rover hardware. Work includes static and impact studies for the URC frame and Ansys structural and flow studies for the MATE rover.

Teaching assistant / AMS 261

Multivariable Calculus, Stony Brook University.

Autodesk ambassador

January 2026 - present. Hosted a Fusion 360, analysis, and rendering workshop for engineering and arts students.

CAD / modeling

SolidWorks, Fusion 360, Inventor, Onshape, Solid Edge, Creo, Siemens NX.

Analysis

Ansys Mechanical, Ansys Fluent, Workbench, LS-DYNA, OpenFOAM, SimScale, OpenRocket.

Software / electronics

Python, C++, KiCad, PyTorch, Qiskit, LaTeX.